# Problem 1

**Game 3**

$$\int_0^\infty e^{-x^{\frac{1}{2025}}}\,\mathrm{d}x$$

The UK University Integration Bee

Chainlink Round

2024/25

Round 3

# Problem 2

**Game 3**

$$\int_{-1}^\infty e^{-x^2}(5x^4-2x^6)\,\mathrm{d}x$$

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# Problem 3

**Game 3**

$$\int (x+2x^2+3x^3+\cdots+2025x^{2025})(1+x+x^2+\cdots+x^{2024})\,\mathrm{d}x$$

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# Problem 4

**Game 3**

$$\int_0^{\frac{\pi}{4}} \frac{\sin(2x)}{\sin(x)+\cos(x)+1}\,\mathrm{d}x$$

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## Problem 5

**Game 3**

$$\int_0^{\frac{\pi}{2}} \ln(\sin(x)) \tan(x) \, \mathrm{d}x$$



## Problem 6

**Game 3**

$$\int_0^{\infty} \frac{x^{\varphi} \tan^{-1}(x)}{1 + x^{2\varphi^2}} \, \mathrm{d}x$$

$$\text{where } \varphi = \frac{1+\sqrt{5}}{2} \text{ is the golden ratio}$$



## Problem 7

**Game 3**

$$\int_0^2 (3x^2 - 5x + 3) \cos(x^3 - 3x^2 + 4x - 2) \, \mathrm{d}x$$



## Problem 8

**Game 3**

$$\int_0^{\frac{\pi}{2}} \frac{1 - \ln(\cos(x))}{x^2 + (1 - \ln(\cos(x)))^2} \cos(2x) \, \mathrm{d}x$$